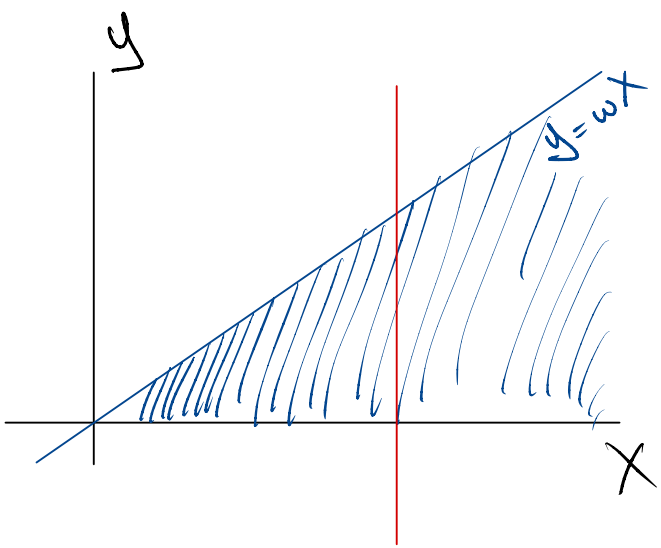
Example-4.16

X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \lambda\mu e^{-(\lambda x+\mu y)} & x \geq 0, y \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Find the PDF of $W = Y/X$. $= F_W(w) = P(W \leq w) \Rightarrow P(Y/X \leq w) = P(Y \leq wX)$



$$\begin{aligned} P[Y \leq wX] &= \int_0^\infty \left(\int_0^{wx} f_{X,Y}(x,y)\, dy \right) dx \\ &= \int_0^\infty \lambda e^{-\lambda x} \left(\int_0^{wx} \mu e^{-\mu y}\, dy \right) dx \\ &= \int_0^\infty \lambda e^{-\lambda x} \left(1 - e^{-\mu w x}\right) dx \\ &= 1 - \frac{\lambda}{\lambda + \mu w}. \end{aligned}$$

PDF of w

CDF of w

Differentiating with respect to w, we obtain

$$f_W(w) = \begin{cases} \dfrac{\lambda\mu}{(\lambda + \mu w)^2} & w \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

Conditional Probability Model

P(Y|X=2): conditional distribution of Y given X = 2

Example: Conditional Probabilities for Mobile Response Time

The response time is the speed of page downloads and it is critical for a mobile Web site. Let X denote the number of bars of service, and let Y denote the response time (to the nearest second) for a particular user and site.

Y=response time (second)	X=number of bars of signal strength		
	1	2	3
4	0.15	0.1	0.05
3	0.02	0.1	0.05
2	0.02	0.03	0.2
1	0.01	0.02	0.25

$= P(X=1) = 0.2$ $\qquad$ $P(X=3) = 0.55$

From the notation for conditional probability in Chapter 1, we can write such conditional probabilities as P(Y = 1 | X = 3) and P(Y = 1 | X = 1).

$$P(y=1 \mid x=3) = \frac{P(y=1, x=3)}{P(x=3)} = \frac{0.25}{0.55} = 0.454$$

$$P(y=1 \mid x=1) = \frac{P(y=1, x=1)}{P(x=1)} = \frac{0.01}{0.2} = 0.05$$

$$P(y=2 \mid x=3) = 0.364$$

$$P(y=3 \mid x=3) = 0.091$$

$$P(y=4 \mid x=3) = 0.091$$

$$P_{y|x=3} = \begin{cases} 0.454 & y=1, x=3 \\ 0.364 & y=2, x=3 \\ 0.091 & y=3, x=3 \\ 0.091 & y=4, x=4 \end{cases}$$

$$= 1$$

Continuous random variables

Conditional Probability Density Function

Given continuous random variables X and Y with joint probability density function $f_{XY}(x, y)$, the **conditional probability density function** of Y given $X = x$ is

$$f_{Y|x}(y) = \frac{f_{XY}(x,y)}{f_X(x)} \quad \text{for} \quad f_X(x) > 0$$

Because the conditional probability density function $f_{Y|X}(y)$ is a probability density function for all y in R_X, the following properties are satisfied:

1. $$f_{Y|x}(y) \geq 0$$
2. $$\int f_{Y|x}(y)\, dy = 1$$

Example: Server Access Time

Let the random variable X denote the time until a computer server connects to your machine (in milliseconds), and let Y denote the time until the server authorizes you as a valid user (in milliseconds). Each of these random variables measures the wait from a common starting time and X < Y. Assume that the joint probability density function for X and Y is

$$f_{XY}(x,y) = 6 \times 10^{-6} \exp(-0.001x - 0.002y) \quad \text{for} \quad x < y$$

$$f_{Y|X}(y) = \frac{f_{XY}(x,y)}{f_X(x)}$$

For the random variables that denote times, determine the conditional probability density function for Y given that X = x.

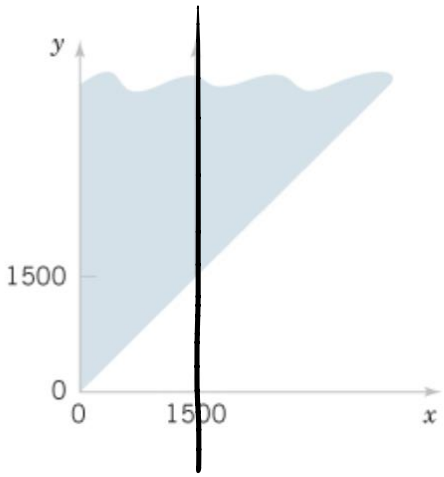
$$\begin{aligned} f_X(x) &= \int_x^{\infty} 6 \times 10^{-6} e^{-0.001x - 0.002y}\, dy \\ &= 6 \times 10^{-6} e^{-0.001x} \left(\frac{e^{-0.002y}}{-0.002} \Big|_x^{\infty} \right) \\ &= 6 \times 10^{-6} e^{-0.001x} \left(\frac{e^{-0.002x}}{0.002} \right) \\ &= 0.003 e^{-0.003x} \quad \text{for} \quad x > 0 \end{aligned}$$

$$\begin{aligned} f_{Y|x}(y) &= \frac{f_{XY}(x,y)}{f_x(x)} = \frac{6 \times 10^{-6} e^{-0.001x - 0.002y}}{0.003 e^{-0.003x}} \\ &= 0.002 e^{0.002x - 0.002y} \quad \text{for} \quad 0 < x \text{ and } x < y \end{aligned}$$

Example: Server Access Time

Determine the probability that Y exceeds 2000, given that x = 1500. That is, determine P(Y > 2000 | X = 1500).

y>x



$$
\begin{aligned}
P(Y > 2000 | X = 1500) &= \int_{2000}^{\infty} f_{Y|1500}(y)\, dy \\
&= \int_{2000}^{\infty} 0.002 e^{0.002(1500) - 0.002y}\, dy \\
&= 0.002 e^{3} \left(\frac{e^{-0.002y}}{-0.002} \Big|_{2000}^{\infty} \right) \\
&= 0.002 e^{3} \left(\frac{e^{-4}}{0.002} \right) = 0.368
\end{aligned}
$$

Conditional Mean and Variance

$E(y) = \int y \cdot f_y(y) dy$

Conditional Mean and Variance

The **conditional mean** of Y given $X = x$, denoted as $E(Y \mid x)$ or $\mu_{Y|x}$, is

$$E(Y|x) = \int_y y\, f_{Y|x}(y)\, dy$$

and the **conditional variance** of Y given $X = x$, denoted as $V(Y \mid x)$ or $\sigma^2_{Y|x}$, is

$$V(Y|x) = \int_y (y - \mu_{Y|x})^2 f_{Y|x}(y)\, dy = \int_y y^2 f_{Y|x}(y)\, dy - \mu^2_{Y|x}$$

$Var(y) = E(y^2) - E(y)^2$

$Var(y|x=x) = E(y^2|x=x) - E(y|x=x)^2$

Example: Server Access Time

E(Y|X = 1500)?

$y > x$

$$E(Y|X=1500) = \int_{1500}^{\infty} y(0.002e^{0.002(1500)-0.002y})\,dy$$

$$= 0.002e^{3}\int_{1500}^{\infty} ye^{-0.002y}\,dy$$

$$\int_{1500}^{\infty} ye^{-0.002y}\,dy = y\frac{e^{-0.002y}}{-0.002}\Big|_{1500}^{\infty} - \int_{1500}^{\infty}\left(\frac{e^{-0.002y}}{-0.002}\right)dy$$

$$= \frac{1500}{0.002}e^{-3} - \left(\frac{e^{-0.002y}}{(-0.002)(-0.002)}\Big|_{1500}^{\infty}\right)$$

$$= \frac{1500}{0.002}e^{-3} + \frac{e^{-3}}{(0.002)(0.002)}$$

$$= \frac{e^{-3}}{0.002}(2000)$$

If the connect time is 1500 ms, then the expected time to be authorized is 0.002(e^3)(2000) ms.

For the discrete random variables in Example: Conditional Probabilities for Mobile Response Time, the conditional mean of Y given X = 3 is obtained from the conditional distribution

Y=response time (second)	X=number of bars of signal strength		
	1	2	3
4	0.15	0.1	0.05
3	0.02	0.1	0.05
2	0.02	0.03	0.2
1	0.01	0.02	0.25

$P(Y=1|X=3) = 0.494$

$P(Y=2|X=3) = 0.364$

$P(Y=3|X=3) = 0.091$

$P(Y=4|X=3) = 0.091$

The conditional mean is interpreted as the expected response time given that three bar of signal is present. The conditional variance of Y given X = 3 is

$$E(Y|X=3) = \sum_{y\in\{1,2,3,4\}} y \cdot P_{Y|X} = 1\times 0.494 + 2\times 0.364 + 3\times 0.091 + 4\times 0.091$$
$$= 1.819$$

$$V(Y|X=3) = E(Y^2|X=3) - E(Y|X=3)^2 = 4.189 - 1.819^2 = \underline{0.876}$$

$$E(Y^2|X=3) = \sum_{y\in\{1,2,3,4\}} y^2 P_{Y|X} = 1^2\times 0.494 + 2^2\times 0.364 + 3^2\times 0.091 + 4^2\times 0.091$$
$$= 4.189$$

Example

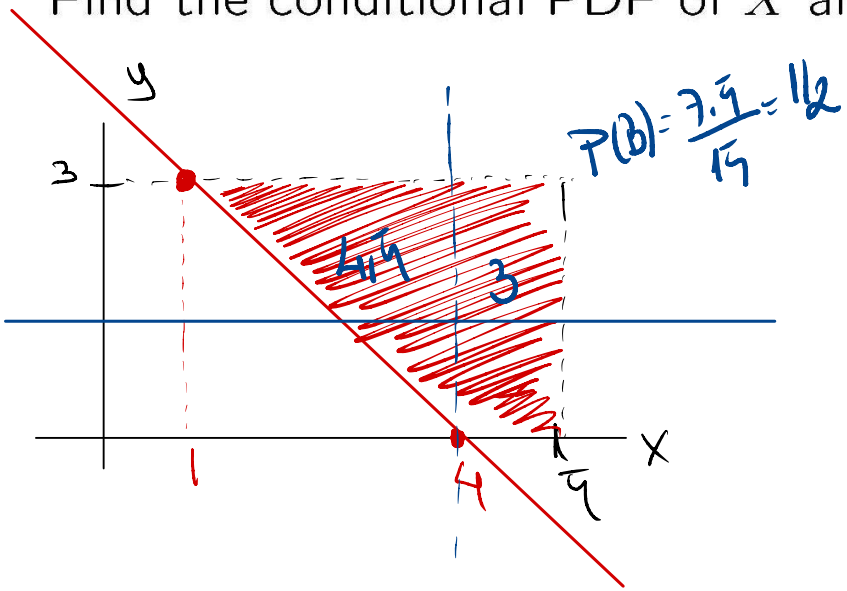
$$f_{XY|X+Y\geq 4} = \frac{f_{XY}(x,y)}{P(X+Y\geq 4)} = \frac{1/15}{1/2}$$

X and Y are random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/15 & 0 \leq x \leq 5, 0 \leq y \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

$$= 2/15 \quad 0 \leq x \leq 5,\ 0 \leq y \leq 3,\ x+y \geq 4$$

Find the conditional PDF of X and Y given the event $B = \{X + Y \geq 4\}$.



$$P(B) = \frac{7.5}{15} = 1/2$$

$$P(B) = P(X+Y\geq 4) = \int_0^3 \int_{4-y}^5 1/15 \, dx\, dy = 1/2$$